

Adam McNelis Mahmoud

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EDUCATION

University of California, Berkeley

B.A. in Data Science — *concentration in Applied Math and Modeling*

B.A. in Applied Mathematics — *concentration in Numerical Analysis*

Coursework: Data Structures, Data Engineering, Machine Learning, Probability, Data Mining, Linear Algebra, Numerical Analysis

Berkeley, CA

Aug. 2022 – May 2026

GPA: 3.86 / 4.00

EXPERIENCE

Data Scientist

June 2026 – Present

Aven

Campbell, CA

- Diagnosed a prolonged decline in click through rate on a Home Equity card upgrade screen by decomposing conversion across customer segments, acquisition sources, and product types, identifying an external partner policy change that reduced transactor traffic and consequently shifted the product mix away from 3% cash back.
- Developed a delinquency-risk screening strategy for renter applicants, optimizing model score thresholds using precision-recall tradeoffs to identify a cutoff that captured 37% of fraudulent accounts while reducing the 30-day delinquency rate from 4.0% to 1.8%.
- Analyzed incomplete balance-transfer behavior, finding that 81% of unswept external balances were attributable to insufficient credit limits and informing targeted credit line increases adopted in production, unlocking \$430K in additional transferable balance.

Data Scientist Intern → Data Scientist

May 2025 – Mar. 2026

Omnium

San Diego, CA → Remote

- Designed ANOVA frameworks and cross-price elasticity regressions to quantify substitution effects across product pack sizes, identifying statistically significant interactions and informing a concurrent ~15% promotion between targeted size pairs.
- Constructed and compared regression-based demand models to quantify market preferences, projecting ~\$9 million sales growth opportunity from new flavor launches and establishing a reusable framework for product innovation analyses.
- Led 15 account-level planning sessions with client representatives and brokers across national retailers, delivering data-driven distribution, pricing, and promotional recommendations resulting in \$1–\$2 retail price increases across dozens of products.

Data Intern

Jan. – May 2025

Fung Institute for Engineering Leadership, UC Berkeley

Berkeley, CA

- Engineered and implemented an end-to-end data pipeline that automates source data ingestion, cleaning, standardization, and reconciliation across academic systems, integrating analyst quality assurance checkpoints and powering real-time metrics and visualizations on the program's live website.
- Conducted outcome and equity analyses on student and alumni data, delivering faculty-facing recommendations that shaped cohort composition, accessibility strategy, and program resource allocation.

Data Analyst

Sept. 2023 – Mar. 2025

Enrollment Management Department, UC Berkeley

Berkeley, CA

- Designed policy-constrained enrollment scenario models in SQL and Python, analyzing admissions yield and student residency tradeoffs to deliver university targets that contributed to enrolling over 700 additional in-state students in 2025.
- Architected and launched the Enrollment Management website, translating institutional data into accessible visualizations and digestible metrics to communicate enrollment and demographic insights to campus stakeholders.

PROJECTS

Equitable College Planner | Python, Streamlit, Git, DuckDB, PCA

UC Berkeley Datathon for Social Good

- Built an interactive college recommendation tool that ranks institutions by student-defined fit using weighted similarity scoring across numeric and categorical features, implemented through a DuckDB-based data pipeline.
- Developed an ROI model incorporating residency-adjusted cost, retention and graduation outcomes, and earnings estimates; visualized personalized recommendations using PCA projections of schools and a user preference vector.

Seasonality Clustering – A Hierarchical Agglomerative Approach | Python, SQL, VS Code

Omnium Internship

- Implemented hierarchical agglomerative clustering on product seasons with custom distance metrics combining velocity variation and temporal gaps to favor continuous seasonal clusters; evaluated models across multiple linkage criteria using dendrograms and silhouette scores.
- Applied clustered season definitions to pricing regressions, reducing multicollinearity compared to standard 13-period models and achieving higher R^2 values with improved demand predictions.

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, MATLAB, HTML/CSS

Libraries: Pandas, NumPy, scikit-learn, statsmodels, SciPy, PyTorch, PyMongo, Keras, Matplotlib/Seaborn

Databases & Tools: PostgreSQL, DuckDB, MongoDB (NoSQL), Excel, Tableau, Git, Jupyter, Snowflake

Methods: Regression, ANOVA, Hypothesis (A/B) Testing, Forecasting, Clustering, Econometrics, PCA, Neural Networks